Role of extracellular vesicles in the pathogenesis of mosquito-borne flaviviruses that impact public health.

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Mosquito-borne flaviviruses represent a public health challenge due to the high-rate endemic infections, severe clinical outcomes, and the potential risk of emerging global outbreaks. Flavivirus disease pathogenesis converges on cellular factors from vectors and hosts, and their interactions are still unclear. Exosomes and microparticles are extracellular vesicles released from cells that mediate the intercellular communication necessary for maintaining homeostasis; however, they have been shown to be involved in disease establishment and progression. This review focuses on the roles of extracellular vesicles in the pathogenesis of mosquito-borne flavivirus diseases: how they contribute to viral cycle completion, cell-to-cell transmission, and cellular responses such as inflammation, immune suppression, and evasion, as well as their potential use as biomarkers or therapeutics (antiviral or vaccines). We highlight the current findings concerning the functionality of extracellular vesicles in different models of dengue virus, Zika virus, yellow fever virus, Japanese encephalitis virus, and West Nile virus infections and diseases. The available evidence suggests that extracellular vesicles mediate diverse functions between hosts, constituting novel effectors for understanding the pathogenic mechanisms of flaviviral diseases.

Climate change and the spread of Aedes mosquito-borne viruses in Europe.

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Several outbreaks of chikungunya and dengue occurred on Mediterranean coasts during the hot season in the last two decades. Aedes albopictus was the vector involved in all the events. As a consequence of climate change, the 'Tiger' mosquito is now spreading through central Europe, and in the summer of 2023, for the first time, mosquito control measures were implemented in Paris to prevent autochthonous transmission of dengue. Rapid changes in the distribution of tropical disease vectors need to be taken into account in future risk assessment activities.

Genetic heterogeneity of the dengue vector Aedes aegypti in Martinique.

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In Martinique, Aedes aegypti, the vector of dengue viruses has been the target of insecticide control for more than 35 years. Despite significant control efforts, dengue has become a major disease of public health concern. We conducted a population genetic analysis based on isoenzyme variations combined with an estimation of infection rate to a dengue virus among 26 Ae. aegypti samples. Aedes aegypti samples could be differentiated for their susceptibility to dengue infection (infection rates ranging from 42.8% to 98.6%) and showed important genetic variation (significant F(ST) values).

Dengue infection increases the locomotor activity of Aedes aegypti females.

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Aedes aegypti is the main vector of the virus causing Dengue fever, a disease that has increased dramatically in importance in recent decades, affecting many tropical and sub-tropical areas of the globe. It is known that viruses and other parasites can potentially alter vector behavior. We investigated whether infection with Dengue virus modifies the behavior of Aedes aegypti females with respect to their activity level. We carried out intrathoracic Dengue 2 virus (DENV-2) infections in Aedes aegypti females and recorded their locomotor activity behavior. We observed an increase of up to ~50% in the activity of infected mosquitoes compared to the uninfected controls. Dengue infection alters mosquito locomotor activity behavior. We speculate that the higher levels of activity observed in infected Aedes aegypti females might involve the circadian clock. Further studies are needed to assess whether this behavioral change could have implications for the dynamics of Dengue virus transmission.

Comments on the epidemiology, pathogenesis and control of dengue.

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Dengue is currently the most important viral disease transmitted to man by arthropods, whether measured by the number of cases or number of deaths. Prevalence of the disease is highest in tropical Asia, intermediate in tropical America, and lowest in tropical Africa. Four distinct dengue viruses have been identified. Types 2 and 3 appear to be more pathogenic on the average than types 1 and 4, but all four can cause severe or fatal dengue syndromes. Infection with any of the four viruses confers life-long homotypic, but not heterotypic, immunity. Dengue viruses can be transmitted by several mosquito species of the genus Aedes, but by far the most common vector is Aedes aegypti. All non-human primates that have been tested are susceptible to infection, but none exhibit signs of illness. The resulting lack of a suitable experimental host other than man has slowed progress in understanding the pathogenesis of severe forms of the disease. Controversy continues over the hypothesis that a first dengue infection increases the risk of severe disease upon re-infection as well as over the principal sites of replication. With regard to replication sites, some authors have mentioned cells of the mononuclear phagocyte lineage while others have implicated hepatocytes. No vaccine is currently available for dengue. Because the principal mosquito vector can use a wide variety of small domestic containers, vector control programs have not been highly effective.

[Dengue fever--not just a tropical infectious disease].

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Dengue fever is a viral disease that is transmitted primarily by Aedes mosquitoes, i. e., A. aegypti and A. albopictus. Other species are rarely involved. The disease is caused by dengue virus, an enveloped RNA virus which belongs to the family of flaviviridae. Although most infections are asymptomatic, in 20 to 30 percentages all cases infections are accompanied with high fever and other influenza-like signs of illness. Serious medical conditions with lethal complications also occur. During the last decades, the incidence of dengue fever rose sharply in many tropical and subtropical countries. In some of these regions, dengue is one of the leading causes of death in children. In Europe, since a few years a strong clustering of dengue fever cases has been registered in travelers returning from certain tropical or subtropical regions. Recently, autochthonous outbreaks have been observed on the Atlantic island of Madeira and in a few other regions of South Europe. Treatment of dengue fever is supportive and symptomatic, a specific therapy does not exist. For prevention of disease, vector control is of crucial importance.

[Vectors of dengue in Mexico: a critical review)].

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The mosquito Aedes (Stegomyia) aegypti (Linneaus, 1762) is the vector of dengue viruses in America. Recently, another related species, Aedes albopictus (Skuse, 1894) was first introduced to America in 1986 in Texas, USA and later on in Sao Paulo, Brazil. Since then, this mosquito has rapidly colonized new areas along the northern border states of Mexico. The importance of this species as a vector of dengue virus is not clear. This work presents a revision of some characteristics of the two species in Mexico, with special emphasis in the documented vector, A. aegypti. Additionally, it includes a critical analysis of the methods for vector surveillance and its importance for a better application of control strategies, as well as some ideas about research needs and priorities in Mexico.

A novel entomological index, Aedes aegypti Breeding Percentage, reveals the geographical spread of the dengue vector in Singapore and serves as a spatial risk indicator for dengue.

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Aedes aegypti is an efficient primary vector of dengue, and has a heterogeneous distribution in Singapore. Aedes albopictus, a poor vector of dengue, is native and ubiquitous on the island. Though dengue risk follows the dispersal of Ae. aegypti, the spatial distribution of the vector is often poorly characterized. Here, based on the ubiquitous presence of Ae. albopictus, we developed a novel entomological index, Ae. aegypti Breeding Percentage (BP), to demonstrate the expansion of Ae. aegypti into new territories that redefined the dengue burden map in Singapore. We also determined the thresholds of BP that render the specific area higher risk of dengue transmission. We performed analysis of dengue fever incidence and Aedes mosquito breeding in Singapore by utilizing island-wide dengue cases and vector surveillance data from 2003 to 2013. The percentage of Ae. aegypti breeding among the total Aedes breeding habitats (BP), and the reported number of dengue fever cases in each year were calculated for each residential grid. The BP of grids, for every year over the 11-year study period, had a consistent positive correlation with the annual case counts. Our findings suggest that the geographical expansion of Ae. aegypti to previously "non-dengue" areas have contributed substantially to the recent dengue fever incidence in Singapore. Our analysis further indicated that non-endemic areas in Singapore are likely to be at risk of dengue fever outbreaks beyond an Ae. aegypti BP of 20%. Our analyses indicate areas with increasing Ae. aegypti BP are likely to become more vulnerable to dengue outbreaks. We propose the usage of Ae. aegypti BP as a factor for spatial risk stratification of dengue fever in endemic countries. The Ae. aegypti BP could be recommended as an indicator for decision making in vector control efforts, and also be used to monitor the geographical expansion of Ae. aegypti.

Dengue infection.

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Dengue is widespread throughout the tropics and local spatial variation in dengue virus transmission is strongly influenced by rainfall, temperature, urbanization and distribution of the principal mosquito vector Aedes aegypti. Currently, endemic dengue virus transmission is reported in the Eastern Mediterranean, American, South-East Asian, Western Pacific and African regions, whereas sporadic local transmission has been reported in Europe and the United States as the result of virus introduction to areas where Ae. aegypti and Aedes albopictus, a secondary vector, occur. The global burden of the disease is not well known, but its epidemiological patterns are alarming for both human health and the global economy. Dengue has been identified as a disease of the future owing to trends toward increased urbanization, scarce water supplies and, possibly, environmental change. According to the WHO, dengue control is technically feasible with coordinated international technical and financial support for national programmes. This Primer provides a general overview on dengue, covering epidemiology, control, disease mechanisms, diagnosis, treatment and research priorities.

Life as a Vector of Dengue Virus: The Antioxidant Strategy of Mosquito Cells to Survive Viral Infection.

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Dengue fever is a mosquito-borne viral disease of increasing global importance. The disease has caused heavy burdens due to frequent outbreaks in tropical and subtropical areas of the world. The dengue virus (DENV) is generally transmitted between human hosts via the bite of a mosquito vector, primarily Aedes aegypti and Ae. albopictus as a minor species. It is known that the virus needs to alternately infect mosquito and human cells. DENV-induced cell death is relevant to the pathogenesis in humans as infected cells undergo apoptosis. In contrast, mosquito cells mostly survive the infection; this allows infected mosquitoes to remain healthy enough to serve as an efficient vector in nature. Overexpression of antioxidant genes such as superoxide dismutase (SOD), catalase (CAT), glutathione peroxidase (GPx), glutathione S-transferase (GST), glutaredoxin (Grx), thioredoxin (Trx), and protein disulfide isomerase (PDI) have been detected in DENV2-infected mosquito cells. Additional antioxidants, including GST, eukaryotic translation initiation factor 5A (eIF5a), and p53 isoform 2 (p53-2), and perhaps some others, are also involved in creating an intracellular environment suitable for cell replication and viral infection. Antiapoptotic effects involving inhibitor of apoptosis (IAP) upregulation and subsequent elevation of caspase-9 and caspase-3 activities also play crucial roles in the ability of mosquito cells to survive DENV infection. This article focused on the effects of intracellular responses in mosquito cells to infection primarily by DENVs. It may provide more information to better understand virus/cell interactions that can possibly elucidate the evolutionary pathway that led to the mosquito becoming a vector.